

**भारतीय मानक ब्यूरो**
BUREAU OF INDIAN STANDARDS

MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG, NEW DELHI 110002

व्यापक परिचालन मसौदा**हमारा संदर्भ: सीईडी 15/टी-47****19/08/2026****तकनीकी समिति : बिल्डर्स हार्डवेयर अनुभागीय समिति , सीईडी 15****प्राप्तकर्ता :**

- क) सिविल इंजीनियरी विभाग परिषद्, सीईडीसी के सभी सदस्य
- ख) बिल्डर्स हार्डवेयर अनुभागीय समिति , सीईडी15 के सभी सदस्य
- ग) रूचि रखने वाले अन्य निकाय

महोदय/महोदया,

निम्नलिखित मानक का मसौदा संलग्न है:

प्रलेख संख्या	शीर्षक
सीईडी 15(34841)WC	पुल हैंडल्स — विशिष्टि का भारतीय मानक मसौदा (IS 17706 का पहला पुनरीक्षण) (आई सी एस संख्या: 91.060.50)

कृपया इस मानक के मसौदे का अवलोकन करें और अपनी समितियों यह बताते हुए भेजे कि यदि यह मानक के रूप में प्रकाशित हो तो इस पर अमल करने में आपके व्यवसाय अथवा कारोबार में क्या कठिनाइयाँ आ सकती हैं ।

समितियों भेजने की अंतिम तिथि : **30 सितम्बर 2026**

समिति यदि कोई हो तो कृपया अधोहस्ताक्षरी को ई-मेल द्वारा ced@bis.gov.in पर या उपरलिखित पते पर, संलग्न फॉर्मेट में भेजें। समितियाँ बीआईएस ई-गवर्नेंस पोर्टल, www.manakonline.in के माध्यम से ऑनलाइन भी भेजी जा सकती हैं।

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धन्यवाद ।

भवदीय,

(द्वैपायन भट्टा.)
प्रमुख (सिविल इंजीनियरी)**संलग्नक : उपरिलिखित**



भारतीय मानक ब्यूरो
BUREAU OF INDIAN STANDARDS

MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG, NEW DELHI 110002

DRAFT IN WIDE CIRCULATION

Our Ref: CED 15/T- 47

19/08/2026

TECHNICAL COMMITTEE: BUILDER'S HARDWARE SECTIONAL COMMITTEE, CED 15

ADDRESSED TO:

- a) All Members of Civil Engineering Division Council, CEDC
- b) All Members of Builder's Hardware Sectional Committee, CED 15 and
- c) All others Interests

Dear Sir(s),

Please find enclosed the following document:

Doc No.	Title
CED 15(34841) WC	<i>Draft Indian Standard</i> Pull Handles — Specification (First Revision of IS 17706) (ICS No. 91.060.50)

Kindly examine the draft standard and forward your views stating any difficulties which you are likely to experience in your business or profession, if this is finally adopted as National Standard.

Last Date for comments: **30th September 2026**

Comments if any, may please be made in the enclosed format and emailed at ced@bis.gov.in or sent at the above address. Additionally, comments may be sent online through the BIS e-governance portal, www.manakonline.in.

In case no comments are received or comments received are of editorial nature, you will kindly permit us to presume your approval for the above document as finalized. However, in case of comments of technical in nature are received then it may be finalized either in consultation with the Chairman, Sectional Committee or referred to the Sectional Committee for further necessary action if so desired by the Chairman, Sectional Committee.

The document is also hosted on BIS website www.bis.gov.in.

Thanking you,

Yours faithfully,

(Dwaipayan Bhadra)

Head (Civil Engg.)

Encl: as above

FORMAT FOR SENDING COMMENTS ON THE DOCUMENT

[Please use A4 size sheet of paper only and type within fields indicated. Comments on each clause/sub-clause/ table/figure, etc, be stated on a fresh row. Information/comments should include reasons for comments, technical references and suggestions for modified wordings of the clause. Comments through e-mail ced@bis.gov.in shall be appreciated.]

Doc. No.: CED 15(34841)WC

BIS Letter Ref: CED 15/T-47

Title: Pull Handles — Specification (*first revision of IS 17706*) (ICS No. 91.060.50)

Last date of comments: 30th September 2026

Name of the Commentator/ Organization: _____

SI No.	Clause/ Para/ Table/ Figure No. commented	Type of Comment (General/ Technical/ Editorial)	Comments/ Modified Wordings	Justification of Proposed Change

NOTE- Kindly insert more rows as necessary for each clause/table, etc

DRAFT FOR COMMENTS ONLY

(Not to be reproduced without the permission of BIS or used as an amendment to Standard)

Builder's Hardware
Sectional Committee, CED 15

Last Date of comments
30th September 2026

Draft Indian Standard

PULL HANDLES — SPECIFICATION
(First Revision)

FOREWORD

(Formal Clauses will be added later)

Pull handles are fittings fixed to doors to help people open and close them easily. They are available in different shapes, sizes and materials and can be used on wooden, metal and glass doors. A good pull handle should be strong, safe and easy to grip, remain firmly fixed during use, and resist damage and corrosion. Where required, pull handles should also be suitable for use with fire and smoke-resistant doors.

This standard was first published in 2022. This first revision has been undertaken to incorporate the experience gained in the implementation of the standard, to update the requirements in line with desired performance requirements of pull handles used on doors in buildings.

The major changes incorporated in this revision are as follows:

- a) The scope and terminology have been updated for more clarity;
- b) Mandatory dimensional requirements for pull handles have been removed to make this standard on performance based approach;
- c) Performance based classification system based on category of use, durability, fire resistance, safety and corrosion resistance has been introduced; and
- d) Requirements for sampling, marking, packing and conformity have been updated.

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed or calculated, expressing the result of a test or analysis shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (second revision)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard

Draft Indian Standard

PULL HANDLES — SPECIFICATION

(First Revision)

1 SCOPE

1.1 This standard specifies the requirements of pull handles fixed to doors in buildings.

1.2 This standard does not cover door and window handles covered under IS 208, IS 4992 respectively and handles designed, manufactured and intended for use in furniture.

2 REFERENCES

2.1 The standards listed in Annex A contain provisions which through reference in the text, constitute provision of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent edition of these standards.

3 TERMINOLOGY

3.1 Gripping length (L_g) — The effective length of a pull handle that can be grasped by hand during operation, measured as the clear distance between the inner faces of the handle supports.

3.2 Pull handles — Handles of any centre-to-centre (c/c) length provided with a back-to-back fixing arrangement or any other fixing arrangement in which the fixings are not visible from the front. Handles with fixings visible from the front shall be classified as pull handles only if they have a gripping length greater than 150 mm.

4 TYPES, DIMENSION AND TOLERANCES OF PULL HANDLES

4.1 Pull handles may be manufactured in different configurations, including D-shape, H-shape, circular, rectangular, square, solid, hollow or other suitable cross-sectional profiles. The clear distance (P_i) from door surface to the inner side of pull handle shall be minimum 30 mm for comfortable grip. The distance (P_o) from door surface to the outer side of pull handle shall be maximum 100 mm for user safety.

4.2 Pull handles generally manufactured with diameter ($\varnothing D$) ranging from 19 mm to 38 mm and length (L) ranging from 150 mm to 2 400 mm. Typical details of D-shape and H-shape pull handles are given in Fig. 1 and Fig. 2 respectively. Where pull handles are manufactured from hollow metal sections, the minimum sheet thickness shall be 0.8 mm.

for diameter ($\varnothing D$) up-to and including 32 mm, and 1.5 mm for diameter ($\varnothing D$) above 32 mm.

4.3 Other shapes and dimensions of pull handles may also be manufactured as per the agreement between manufacturer and purchaser, provided they conform to the performance requirements specified in this standard.

4.4 Tolerances

4.4.1 Tolerances to the dimensions (other than thickness of sheet used) of pull handles shall be ± 1 mm for all sizes and types of pull handles.

4.4.2 For thickness, tolerances shall be as given in the raw material Indian Standards. In case of non-availability tolerance in the raw material Indian Standard, tolerance shall be ± 0.2 mm.

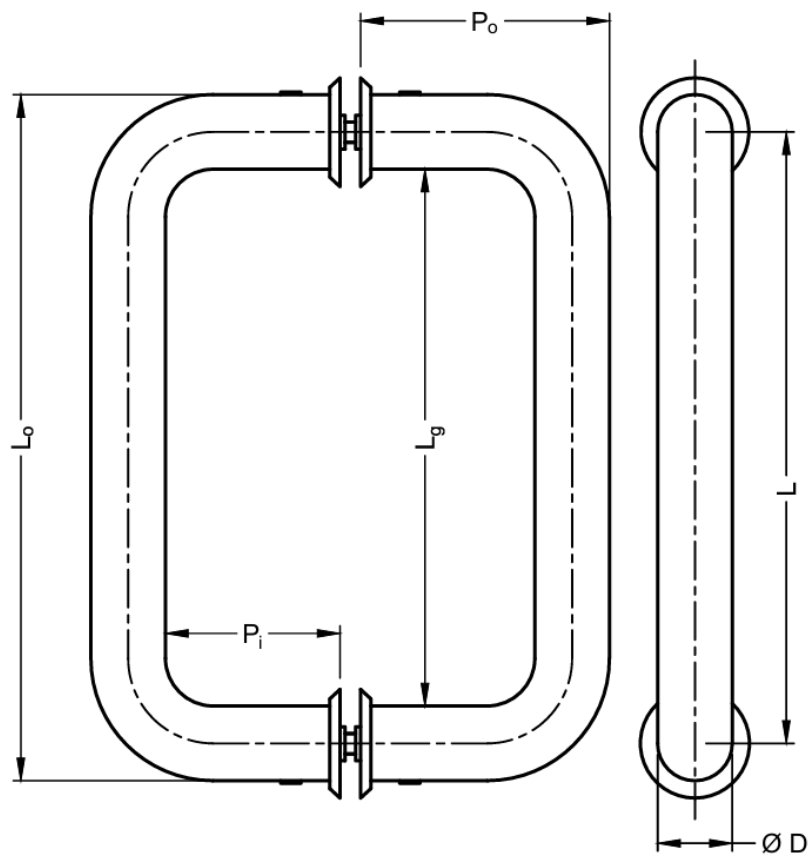


FIG. 1 ILLUSTRATION OF D-SHAPE PULL HANDLE

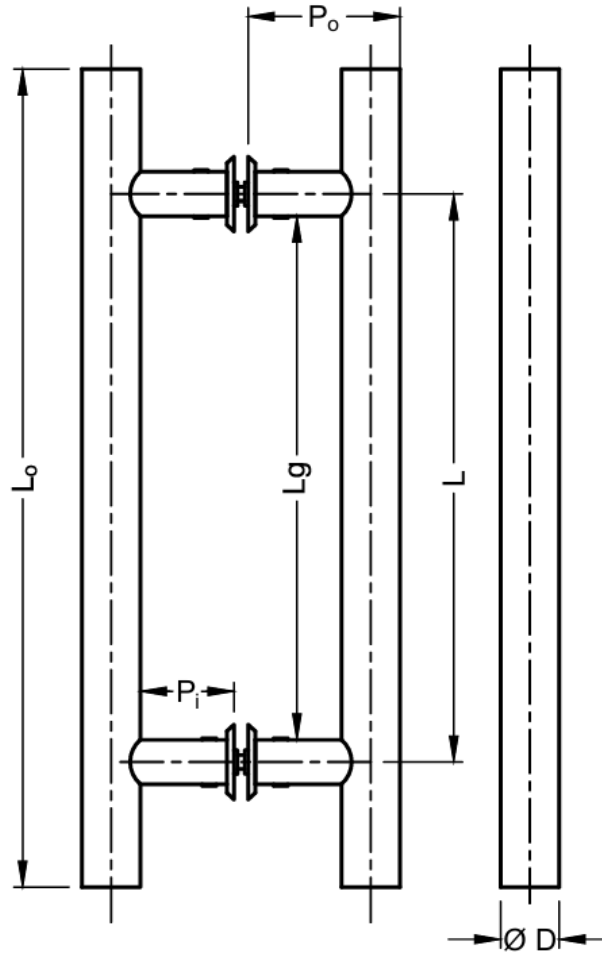


FIG. 2 ILLUSTRATION OF H-SHAPE PULL HANDLE

5 MANUFACTURE

5.1 Materials

Pull handles may be from any suitable material such as steel, stainless steel, brass, bronze, aluminium, zinc alloys, plastics or their combinations. The material selected shall be such that it does not affect the quality and performance of the pull handle as specified in this standard.

5.2 Fitting Accessories and Installation

Pull handles shall be designed for ease of installation and shall be suitable for fixing on doors such as wooden, metal and glass doors, as applicable. All necessary fittings and accessories required for proper installation of the pull handle shall be supplied along with

each door handle by the manufacturer. The manufacturer shall also provide appropriate installation instructions.

5.3 Finish

5.3.1 The pull handles shall be finished with appropriate coating or plating so as to meet the performance and corrosion resistance requirements specified in this standard.

6 CLASSIFICATION CODE

For the purpose of this standard, the door handle shall be classified according to following digit coding system.

6.1 Digit 1 – Category of Use

Based on the intended usage and load carrying capacity, pull handles shall be classified into the following grades:

- a) Grade 1 — Light duty, for handles capable of withstanding a force of 100 N;
- b) Grade 2 — Medium duty, for handles capable of withstanding a force of 250 N;
- c) Grade 3 — Heavy duty, for handles capable of withstanding a force of 600 N; and
- d) Grade 4 — Severe duty, for handles capable of withstanding a force of 1 200 N.

6.2 Digit 2 – Durability

The durability of a pull handle shall be classified as:

- a) Grade 2 — 5 000 test cycles.

6.3 Digit 3 – Door mass

Door mass is not applicable for the classification of pull handles under this standard.

6.4 Digit 4 – Fire Resistance

Two grades are classified for the requirement of use of pull handles on door assemblies are as follows:

- a) Grade 0 — Not suitable for use on fire/smoke check doors; and
- b) Grade 1 — Suitable for use on fire/smoke check doors.

6.5 Digit 5 – Safety

All pull handles are required to satisfy the essential requirements of safety in use. Therefore, only Grade 1 is identified.

6.6 Digit 6 — Corrosion Resistance

6.6.1 In case of pull handle made of ferrous content materials, corrosion resistance is categorized in four grades based on following duration when tested for neutral salt spray test as per IS 5528.

- a) Grade 1 — Low corrosion resistance, up to 24 h;
- b) Grade 2 — Moderate corrosion resistance, up to 48 h;
- c) Grade 3 — High corrosion resistance, up to 96; and
- d) Grade 4 — Very high corrosion resistance, up to 240 h.

6.6.2 In case pull handle made from non-ferrous materials, the corrosion resistance test is not applicable for the classification of pull handles under this standard.

6.7 Examples of Classification

Pull handle intended for medium duty application, having a minimum durability of 5 000 cycles, not suitable for use on fire/smoke check doors, meeting the safety requirements and having moderate corrosion resistance.

Digit 1	Digit 2	Digit 3	Digit 4	Digit 5	Digit 6
2	2	NA	0	1	2

Pull handle intended for severe duty application, having a minimum durability of 5 000 operating cycles, suitable for use on fire/smoke check doors, meeting the safety requirements and made from non-ferrous material.

Digit 1	Digit 2	Digit 3	Digit 4	Digit 5	Digit 6
4	2	NA	1	1	NA

7 PERFORMANCE REQUIREMENTS

7.1 General

When tested as per this standard, the pull handle shall meet the performance requirements given in **7.2** to **7.5** as per the declared classification code (see **6**) of the door handle.

7.2 Durability

7.2.1 The pull handle shall be capable of withstanding the durability cycle as specified in the declared classification code (see **6**). The test shall be carried out by applying the specified operating force corresponding to the category of use.

7.2.2 At the end of the test, the pull handle shall not show:

- a) the pull handle assembly shall not show any sign of damage;
- b) the pull handle fixings shall not show any sign of loosening or damage; and
- c) any residual deformation of the pull handle, as measured in accordance with Annex B, shall be not greater than 1 mm.

The test set-up and method for durability testing are given in Annex B.

7.2.3 The durability test shall be carried out once in six months or in case of any change in shape, size or material.

7.3 Fire Resistance

7.3.1 This requirement is based on the evaluation of the product on test door assemblies as per IS 3614 for 120 minutes. The manufacturer shall indicate the method of fixing which will be used, together with the details of any additional material necessary to achieve the required performance under this test.

7.3.2 The pull handle, when tested as part of the fire door assembly, shall not adversely affect the integrity or performance of the assembly. The product shall conform to the classification given in **6.4**.

7.4 Safety

7.4.1 Pull handles shall be well made, free from defects and shall have smooth finish. All arris and exposed corners shall be rounded to a radius of not less than 0.5 mm to ensure safe and easy handling. Cast handles shall be free from casting defects.

7.4.2 In case of bent pull handles, excessive creasing i.e. sharp wrinkles, folds, or ripples on the inside curve of the pull handle's bend, and necking i.e. reduction in thickness on the outside curve in the bend area shall not be permitted.

7.5 Corrosion Resistance

When tested for neutral salt spray test as per IS 5528 for the duration as per classification code (see **6.6**) grade declared by the manufacturer. After testing, the pull handles shall not show any corrosion on the visible surfaces.

8 SAMPLING AND CRITERION FOR CONFORMITY

8.1 Lot

In any consignment, all the pull handles of the same shape, size and finish, manufactured from similar materials under identical conditions of manufacture, shall be grouped together to constitute a lot.

8.2 Sampling

8.2.1 Pull handles shall be selected from a lot at random. In order to ensure the randomness of selection, the procedure given in IS 4905 shall be followed.

8.2.2 The minimum quantity of pull handles samples to be taken for the each of the testing shall be as follows:

- a) One pull handle for general requirements (workmanship, finish and safety), functional check and operation at extreme temperatures;
- b) One pull handle for durability;
- c) One pull handle for corrosion resistance; and
- d) One pull handle for fire resistance, on test door assemblies as per IS 3614.

8.2.3 The samples of pull handles selected for testing for conformity, shall meet the requirements given in this standard.

8.2.4 Retesting is not required where only aesthetic changes i.e change in visual style without change in c/c length, diameter, thickness, projections, shape, number of joint(s) and material, are made which do not affect the performance characteristics of the pull handle. Where only the coating or surface finish is changed, only the corrosion resistance test, shall be repeated.

9 MARKING

9.1 Each pull handle shall be legibly and indelibly marked with the following information:

- a) Manufacturer's logo or brand name or trade-mark; and
- b) Classification code (see 6).

9.2 Pull handles shall be suitably packed in cartons. Each carton shall be legibly and indelibly marked with the following:

- a) Manufacturer's name or brand name or trade-mark;
- b) Material and finish of door handle;
- c) Type and size [Length (c/c) and overall Length] of pull handle;
- d) Classification code;
- e) In the case of pull handles manufactured from hollow metal sections, the final product metal thickness;
- f) Quantity in the package; and
- g) Batch/lot number.

9.3 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the Bureau of Indian Standards Act, 2016 and the Rules and Regulations framed thereunder, and the products may be marked with the Standard Mark.

10 PACKING

The pull handles shall be packed in cardboard boxes or in any other packing which avoids damages to the product.

ANNEX A

(Clause 2.1)

LIST OF REFERRED INDIAN STANDARDS

<i>IS. No.</i>	<i>Title</i>
IS 208: 2020	Door handles – Specification (<i>sixth revision</i>)
IS 4992: 2024	Door handles for mortice locks — Specification (<i>second revision</i>)
IS 4905: 2015/ISO 24153: 2009	Random sampling and randomization procedures (<i>first revision</i>)
IS 5528: 2024/ ISO 9227: 2022	Corrosion tests in artificial atmospheres — Salt spray tests (<i>second revision</i>)

ANNEX B

(Clause 7.2.2)

METHOD OF DURABILITY TEST

B-1 GENERAL

B-1.1 Control the ambient temperature of the test environment throughout the test from 15°C to 30°C. Ensure that the testing environment is draught free.

B-1.2 The following tolerances on measured values, except where otherwise defined shall be applicable:

- a) angular position, in degrees : $\pm 2^\circ$
- b) force, in newtons : ± 2 percent
- c) length, in millimetres : ± 2 percent
- d) time, in seconds : ± 1 s

B-1.3 Determine measurements using measuring instruments with an accuracy of ≥ 1.5 percent. Include, as part of the test equipment, a means for recording the number of test cycles performed.

B-2 APPARATUS

B-2.1 The test apparatus shall comprise a mounting block and ram as shown in Fig. 3.

B-2.2 Mounting block shall be of plywood made with overall thickness of approximately 40 mm. The mounting block shall be of a suitable size to accept the test sample. The block shall extend at least 30 mm beyond the extreme dimensions of the sample in all

directions, including any mounting plate or fixture that forms a part of the pull handle assembly.

B-2.3 Ram, comprising a hydraulic load applicator, or similar loading device, which can exert the test forces up to required category of application. Three pull action rams on the test rig shall be connected in such a way that each can be operated independently. The pull action rams shall be capable of operating at 3 s intervals. An additional angular positioning ram shall be positioned to enable the three pull action rams to be rotated from a 30° position to a 150° position, in increments of 30°.

B-3 PROCEDURE

B-3.1 Preparing the Test Specimen

Mount the test specimen on the test rig using the fixings with which the manufacturer intends the pull handle to be used.

NOTE — The pull action rams of the test rig are not attached to the test specimen at this stage of the durability.

B-3.2 Pre-test Measurements

Measure the gap between the inner face of the pull handle and the face of the test block, at the mid-point of the handle.

B-3.3 Method of Force Application

B-3.3.1 General

- a) Select the force that a pull action ram exerts on the test specimen as per the category declared.
- b) Connect the pull action ram(s) to the test specimen and subject the test specimen to the force in accordance with B-3.3.2 or B-3.3.3, whichever is applicable.

B-3.3.2 Method of Test for Pull Handles < 200 mm in Length

Connect the middle pull action ram to the centre of the pull handle test specimen. The two outer pull action rams are not connected to the test specimen. When it is not possible to connect the middle pull action ram to the centre of the pull handle, for example pull handles with one central fixing point, connect the middle pull action ram to either side of the fixing point in such a way that the load applied by the ram is evenly distributed and avoids excess stress. In such cases, record the precise position of the middle pull action ram.

Index the angular positioning ram to the 30°, 60°, 90°, 120° and 150° position and apply a force without shock according to the required category of use. Maintain the force for 3 s. Repeat the test cycle 5 000 times.

B-3.3.3 Method of Test for Pull Handles >200 mm in Length

Connect the three pull action rams to the pull handle and ensure that the two outer pull action rams are positioned 50 mm from each end, and the remaining pull action ram is centrally positioned on the handle.

When it is not possible to connect the two outer pull action rams 50 mm from each end of the pull handle, for example for cranked pull handles, ensure that the two outer pull action rams are connected to the next nearest practicable position. In such cases, record the precise position of the outer two pull action rams.

When it is not possible to connect the middle pull action ram to the centre of the pull handle, for example for pull handles with one central fixing point, connect the middle pull action ram to either side of the fixing point in such a way that the load applied by the ram is evenly distributed and avoids excess stress. In such cases, record the precise position of the middle pull action ram.

Index the angular positioning ram to the 30°, 60°, 90°, 120° and 150° position and apply a force without shock according to the required category of use, each of the three pull action rams applying the force in sequence. Maintain the force from each pull action ram for 3 s. Repeat the test cycle 5 000 times.

B-3.3.4 Post-test Measurement

Measure the gap between the inner face of the pull handle and the face of the test block, at the mid-point of the handle.

B-4 CALCULATION OF RESULTS

Determine the extent of deformation caused on the pull handle by calculating the difference between the values determined.

B-5 POST-TEST ASSESSMENT

After the testing, check the sample for the following points:

- a) The pull handle assembly shall not show any sign of damage;
- b) The pull handle fixings shall not show any sign of loosening or damage; and
- c) The deformation of the pull handle shall be not greater than 1 mm.

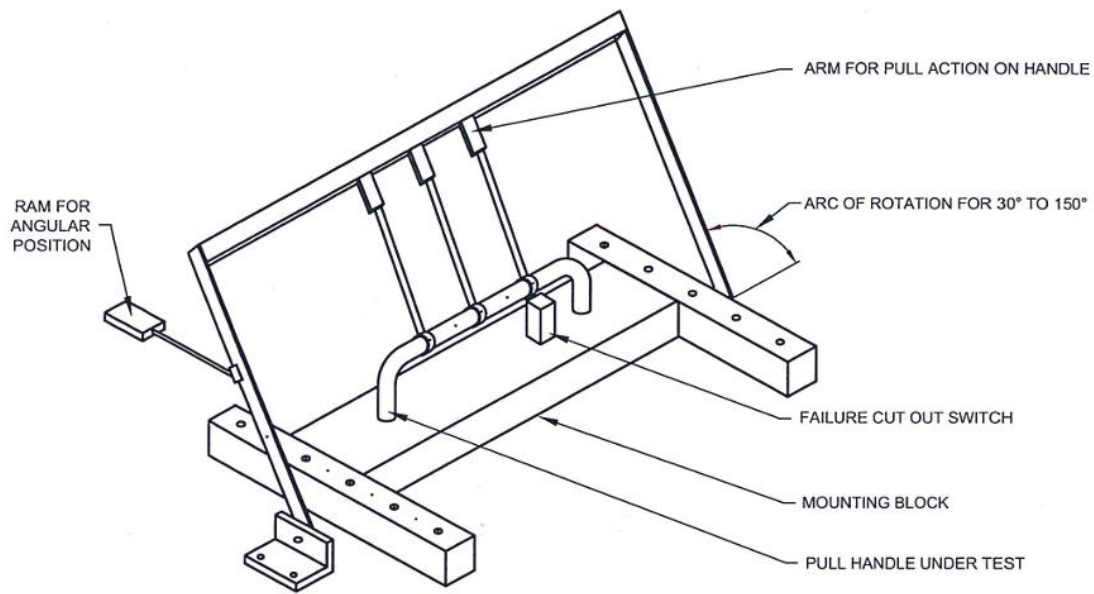


FIG. 3 TYPICAL TEST APPARATUS FOR DURABILITY TEST
